AERA Proposal

Diego Morgado Innocencio
Duncan Lord
Mary Frances Early College of Education
EDIT9990 Doctoral Topical Seminar
Dr. Ai-Chu Elisha Ding
May 21, 2024

Designing an Immersive VR Intervention to Educate Attention and Intention in Pre-Service Music Teachers

Abstract: This study explores how an Immersive Virtual Reality (IVR) intervention, grounded in a 4E cognitive framework and an enactive approach to education, can enhance the attention and intention of pre-service music teachers. The Skilled Intentionality Framework (SIF) is used to explain expertise and "skilled action." The intervention allowed a learner to practice noticing and responding to a simulated ensemble, and included designed scaffolds to support the participant's noticing. The intervention was delivered to a pre-service music teacher, and data was collected through video and audio recording. The results showed changes in the participant's feedback structure over time, with an increase in specificity and greater "attunement" to the feedback given by an expert, indicating the potential benefits of the enactive approach.

Potential Divisions or SIGs: Technology, Instruction, Cognition, and Learning (SIG), Teaching and Teacher Education (K), Music Education (SIG)

Authors: Duncan Lord (59%), Diego Innocencio (39%), Microsoft Copilot (2% - Revising for word count)

Introduction

Pre-service music teachers can sometimes struggle to respond appropriately in their rehearsal settings because the environment is stimulus rich; it can be challenging to identify opportunities in the environment. Further, novices sometimes lack the tools and strategies to respond with nuance and finesse. We perceive this as a knowledge gap between novice and expert music teachers. While this gap can be addressed through teaching experiences, access to quality practicum locations is not always available for all pre-service music teachers. This is especially true at early stages of a preparation program.

Immersive Virtual Reality (IVR) allows learners to practice these skills in a simulated authentic environment with easier access and reduced risk. We created an IVR intervention and conducted this study to learn more about how an IVR intervention might help pre-service music teachers to identify opportunities in the rehearsal environment, and to respond to those opportunities skillfully.

The purpose of this study is to determine how an IVR intervention design, rooted in a 4E cognitive framework, could affect the attentions and intentions of a pre-service music teacher. Our research questions were:

1. How does the participant embody the role of the "director" during the IVR intervention?
2. How does the participant's rehearsal feedback change over time?

Theoretical Framework

This study focuses on an enactive approach to pre-service music teacher education. The enactive approach is rooted in ecological psychology and is also a branch of 4E cognition (Churchill, 2022; Jacobs & Michaels, 2007). The ecological/enactive approach views learning theories as "mere change" theories in which the learner becomes more fit to their environment (Jacobs & Michaels, 2007) and is, therefore, better equipped to respond to that environment in an expert way. This "fitness" is what defines expertise from the enactive/ecological approach (Jacobs & Michaels, 2007).

Another defining aspect of the enactive/ecological approach is the view that learning focuses on the education of *attention* and *intention* (Churchill, 2022; Mason, 2010; Young, 2004). *Attention* refers to the affordances recognized by a person within an environment (Rietveld, 2008; Rietveld et al., 2018). The *education of attention* is the process by which a person comes to recognize more effective or appropriate affordances within an environment (Bruinenburg & Rietveld, 2014; Rietveld, 2008; Rietveld et al., 2018; van Dijk & Rietveld, 2017). *Intention* refers to goals or actions a perceiver plans to apply to recognized affordances in the environment (Jacobs & Michaels, 2007; Young, 2004). We argue that the *education of intention*, then, is the refinement of a person's goals and actions to be more in line with those planned and taken by experts. These two "educations" align with literature on enskilment toward expertise (Churchill, 2022; Rietveld, 2008; Rietveld and Kiverstein, 2014; Woods et al., 2021).

The Skilled Intentionality Framework (SIF) provides another framework for understanding expertise, skilled action, and enskilment within an enactive and 4E cognitive perspective. *Skilled intentionality* is defined as "The kind of intentionality an individual exhibits when acting skillfully in a familiar situation" (Bruinenburg & Rietveld, 2014, p. 2). The SIF is described as a "self-regulating" model that illustrates how a skilled person orchestrates between two loops: their internal (bodily) action readiness, and their

external perceptions of the "rich landscape of affordances" (Rietveld et al., 2018). These two loops of the model (shown in Figure 1) can be understood as mapping onto the previously described notions of intention (bodily action readiness) and attention (perception of the rich landscape of affordances).

The SIF specifically, and the enactive/ecological approach broadly, bring potentially unique perspectives and emphases to the design and implementation of learning experiences for pre-service music teachers. This theoretical framework suggests that the education of both attention and intention would need to be explicitly considered when seeking to advance a novice (pre-service) music teacher toward expertise (taking skilled action) in the classroom. Our study seeks to explore how a design built with this framework in mind could affect the learner's ability to identify and respond to their environment skillfully.

The decision to use IVR as a platform for the intervention is supported by the DICE framework (Bailenson, 2018). The DICE framework suggests that VR is useful for training and education situations which are dangerous, impossible, counterintuitive, or expensive. An IVR instructional intervention would allow pre-service music teachers to be immersed in an authentic and stimulus-rich environment without the need to dedicate time and resources to secure and travel to a real classroom and allows the learner to develop skills without interrupting the learning of real students or being placed in an environment where real students could respond negatively.

Methods

We developed a custom IVR intervention to teach pre-service music teachers how to identify and respond to their rehearsal environment. This custom intervention is a 13 minute, stereoscopic, 180-degree VR video recording of a music rehearsal frame (Duke, 1994) from the perspective of the director. The use of video recordings of real rehearsals allowed us to easily capture and immerse the learner in an authentic rehearsal environment. We were also able to edit the videos to enable instructional activities and educational scaffolds that are not normally possible in a real teaching environment (i.e., pausing, rewinding, visual cues, live teacher commentary), as described below.

The footage contains 3 performance episodes (Duke, 1994) within the same rehearsal frame (Figure 2). Each episode is itself edited and presented in 3 distinct chunks (Figure 3). The first chunk immerses the learner in the authentic rehearsal environment with no additional scaffolds and asks the participant to practice giving feedback to the simulated ensemble. The second chunk repeats the performance with visual cues used to help direct the participant's attention toward possible affordances in the environment which they could have addressed through feedback. The third chunk repeats the performance a final time with commentary provided by the director. This allows the expert director a chance to explain what they noticed and how they decided to approach the rehearsal situation. This final chunk is designed to align with a cognitive apprenticeship approach to teaching (Collins et al., 1987). Each of the three chunks in the performance episode include moments where the video asks the participant to pause and answer questions or participate in practice activities. Figures 4-6 provide visual diagrams of the structure of each individual chunk. Bear in mind that there are 3 different episodes presented in the intervention since the rehearsal frame included 3 performance episodes.

The instructional intervention was delivered using a Meta Quest 2 HMD. An HD webcam and an external USB microphone captured 3rd person footage and spoken audio respectively. A laptop stored the recordings and data. Google Chrome facilitated a screencast of the participant's 1st person experience. Audacity captured spoken dialogue, while Open Broadcasting Software (OBS) captured 1st and 3rd

person video streams separately. DeoVR, installed on the HMD, was used as a video player. Figure 7 illustrates the setup of the collection setting.

The intervention was delivered to a single male, undergraduate, pre-service music teacher from a university in the South-eastern United States. Data was gathered through video and audio recording. Adobe Premiere Pro was used to stitch together and sync the 1st and 3rd person video streams with the spoken audio, and to generate timestamped audio transcripts. The video collage, synced audio, and transcripts were imported into Atlas.ti for qualitative analysis.

Coding

We used some of the branches and concepts from 4E cognition to organize our coding process and findings; a method that has been used by other researchers in music education literature (Paschalidou, 2024). *Figure 8* includes our full coding scheme and memo definitions for these codes. First, we coded overall segments of the footage as being *attentional* or *intentional.*

Research question #1 is concerned with the participant's embodiment, so we created codes that were specific to embodied actions taken by the participant. We categorized actions as being *attentional* embodied actions and *intentional* embodied actions.

Research question #2 is concerned with the participant's feedback, so we coded the speech provided by the participant. We first noticed that the verbal feedback fell into two categories: *embodied* speech and *non-embodied* speech.

Attention and intention are important parts of the theoretical framework, so we wanted to continue delineating between these two with speech as well. So, we applied those codes to the two categories of speech. This resulted in 4 speech codes: *Attentional embodied, attentional non-embodied, intentional embodied, and intentional non-embodied.*

Lastly, we coded the quality of the feedback provided by the participant to the simulated ensemble. *General* feedback dealt with general, global issues present in the ensemble while *specific* feedback was more targeted toward a particular ensemble member, ensemble section, and/or musical segment.

Results

<u>R1. How does the participant embody the role of the "director" during the IVR intervention?</u>

In the IVR intervention, the participant embodied the "director" role, both implicitly and explicitly. Explicit embodiment occurred during *designed moments of embodiment*, where the participant was asked to act as a director and provide feedback. This involved changes in voice, posture, diction, and communication style, with the participant speaking directly to the ensemble in second person. In contrast, other tasks saw a more casual approach, using less technical language and first and third person.

Implicit embodiment was observed throughout the intervention, with gestures like foot tapping, singing, and conducting aiding understanding. These attentional embodied actions were distinct from intentional actions directed at the ensemble. During *designed moments of embodiment*, the participant focused on specific individuals, pairing gestures with verbal feedback for reinforcement. For instance, he

demonstrated a bell swing or conducted and sang to suggest improvements. This study highlights the nuanced ways in which the participant engaged with the role of director during the intervention.

R2. How does the participant's rehearsal feedback change over time?

The participant's feedback provided to the simulated ensemble changed over the course of the intervention. In the initial performance tasks, the participant failed to notice many of the errors and issues that were present in the rehearsal segment. The feedback offered by the participant was, consequently, very general. As the intervention progressed, the feedback became more specific and better directed toward individual ensemble members or musical segments. In each episode, the participant had opportunities to compare his noticing and feedback to an expert's. We found that as the intervention progressed, the participant's feedback became more closely aligned with that of the expert's. This is not to say that his feedback was identical in content, structure, or rehearsal style, but the participant did advance from offering only general feedback to offering a mix of general and specific comments which included some of the issues present in the videos. This small progress was expected, and is consistent with literature dealing with cognitive apprenticeships, attunement, calibration, and enskilment (Churchill, 2022; Collins et al., 1987; Rietveld, 2008; Rietveld & Kiverstein, 2014; Woods et al., 202).

Discussion *(significance, implications, future directions)*

This study extends the SIF and enactive/ecological approach into higher cognition and learning, demonstrating its implementation in creating effective learning interventions. It underscores IVR's potential as a platform for delivering authentic, situated learning experiences. The results suggest that activities supporting the education of attention and intention could benefit pre-service music teachers, reinforcing existing literature on IVR's use to improve teachers' classroom behaviors (Chen, 2022).

The study also provides directions for future progress and studies. A follow-up interview revealed positive sentiment towards the intervention, but also highlighted areas for improvement, such as incorporating access to the music score and addressing the tiring nature of the hour-long intervention without breaks from the headset.

However, the study has limitations. It represents our interpretation of a single participant's experience with a single handbell ensemble rehearsal, limiting generalizability. The participant's familiarity with the ensemble also influenced their interaction and response. Future studies could involve different participants and footage of different ensembles at various age groups, exposing pre-service teachers to diverse teaching situations.

References

Bailenson, J. (2018). *Experience on demand: What Virtual Reality Is, How it Works, and What it can Do* (1st ed.). W.W. Norton & Company.

Bruineberg, J., & Rietveld, E. (2014). Self-organization, free energy minimization, and optimal grip on a field of affordances. *Frontiers in Human Neuroscience*, *8*(AUG). https://doi.org/10.3389/fnhum.2014.00599

Chen, C. Y. (2022). Immersive virtual reality to train preservice teachers in managing students' challenging behaviours: A pilot study. *British Journal of Educational Technology*, *53*(4), 998–1024. https://doi.org/10.1111/bjet.13181

Churchill, E. (2022). The education of attention in enskillment. *Journal of Pragmatics*, *193*, 155–172. https://doi.org/10.1016/J.PRAGMA.2022.03.006

Collins, A., Brown, J. S., & Newman, S. E. (1987). *Technical Report No. 403. Cognitive Apprenticeship: Teaching the Craft of Reading, Writing, and Mathematics.* University of Illinois at Urbana-Champaign Center for the Study of Reading; Bolt, Beranek and Newman, Inc., Cambridge, MA. https://ocw.metu.edu.tr/pluginfile.php/9107/mod_resource/content/1/Collins%20report.pdf

Duke, R. A. (1994). Bringing The Art of Rehearsing Into Focus - The Rehearsal Frame As A Model For Prescriptive Analysis of Rehearsal Conducting. *Journal of Band Research*, *30*(1), 78–95.

Jacobs, D. M., & Michaels, C. R. (2007). Direct learning. *Ecological Psychology*, *19*(4), 321–349. https://doi.org/10.1080/10407410701432337

Mason, J. (2010). Attention and Intention in Learning About Teaching Through Teaching. *Learning Through Teaching Mathematics*, 23–47. https://doi.org/10.1007/978-90-481-3990-3_2

Paschalidou, S. (2024). Technology-Mediated Hindustani Dhrupad Music Education: An Ethnographic Contribution to the 4E Cognition Perspective. *Education Sciences*, 14(2). https://doi.org/10.3390/educsci14020203

Rietveld, E. (2008). Situated Normativity: the normative aspect of embodied cognition in unreflective action. *Mind,* 117 (973–1001). https://doi.org/10.1093/mind/fzn050

Rietveld, E., Denys, D., and van Westen, M. (2018). Ecological- Enactive Cognition as engaging with a field of relevant affordances: The Skilled Intentionality Framework (SIF). In A. Newen, L. de Bruin, & s. Gallagher (Eds.), *The Oxford Handbook of 4E Cognition* (pp. 41- 70). Oxford University Press.

Rietveld, E., & Kiverstein, J. (2014). A rich landscape of affordances. *Ecol. Psychol.* 26, 325–352. https://doi.org/10.1080/10407413.2014.958035

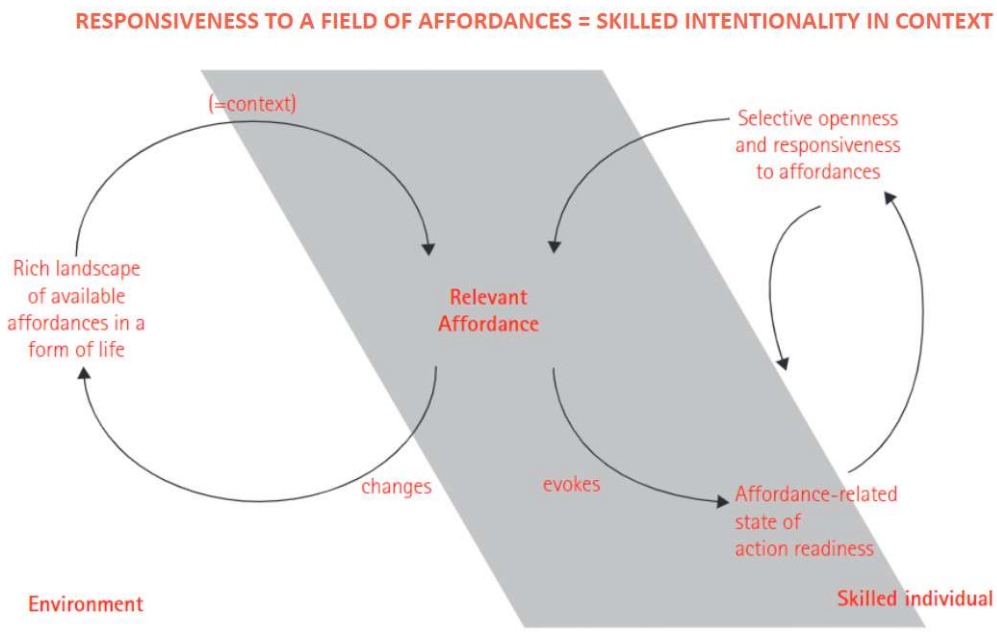
van Dijk, L., & Rietveld, E. (2017). Foregrounding sociomaterial practice in our understanding of affordances: The skilled intentionality framework. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2016.01969

Woods, C.T., Rudd, J., Gray, R., & Davids, K. (2021). Enskilment: an Ecological-Anthropological Worldview of Skill, Learning and Education in Sport. *Sports Med - Open* 7, 33. https://doi.org/10.1186/s40798-021-00326-6

Young, M. (2004). An Ecological Psychology of Instructional Design: Learning and Thinking by Perceiving-Acting Systems. In D. H. Jonassen (Ed.), *Handbook of research on educational communications and technology., 2nd ed*. (pp. 169–177). Lawrence Erlbaum Associates Publishers.

Figures

Figure 1 - The Skilled Intentionality Framework (SIF)



(Bruineberg & Rietveld, 2014)

Figure 2 – Timeline of the Entire Instructional Intervention

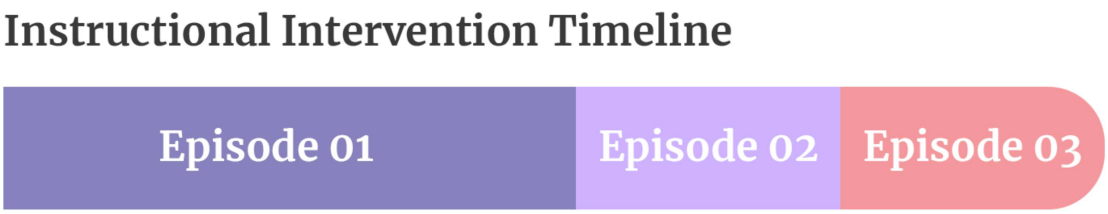


Figure 3 – Anatomy of Each Episode

Episode Anatomy	
Icon	
	Chunk 01
	Pop-up information
	Video watching
	Pop-up information and pause for activity
	Video watching
	Pop-up information and pause for activity
	Chunk 02
	Pop-up information
	Video watching with visual cues, the participant will provide feedback out loud
	Pop-up information and pause for activity
	Chunk 03
	Pop-up information
	Video watching with visual cues and director's commentary
	Pop-up information and pause for activity

Figure 4 – Chunk 1 Visual Diagram

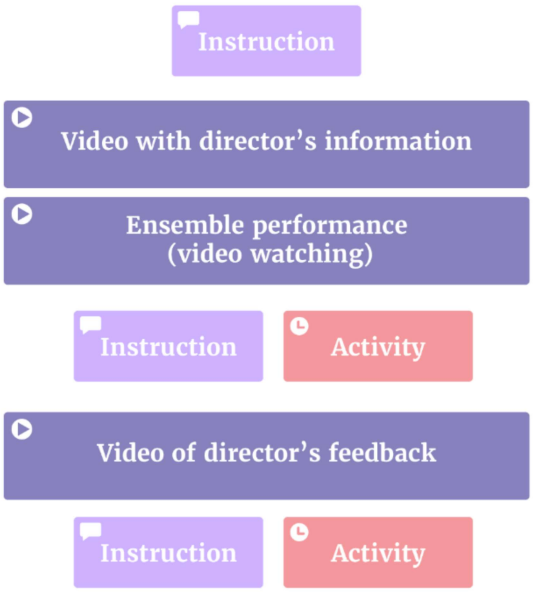


Figure 5 – Chunk 2 Visual Diagram

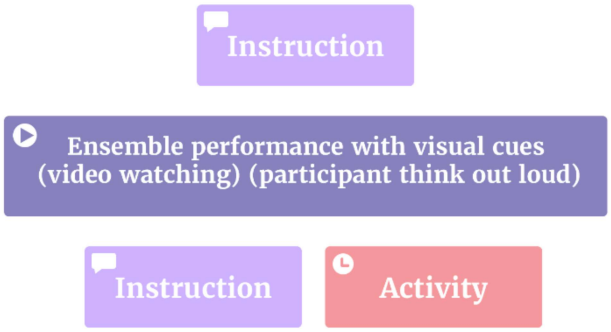


Figure 6 – Chunk 3 Visual Diagram

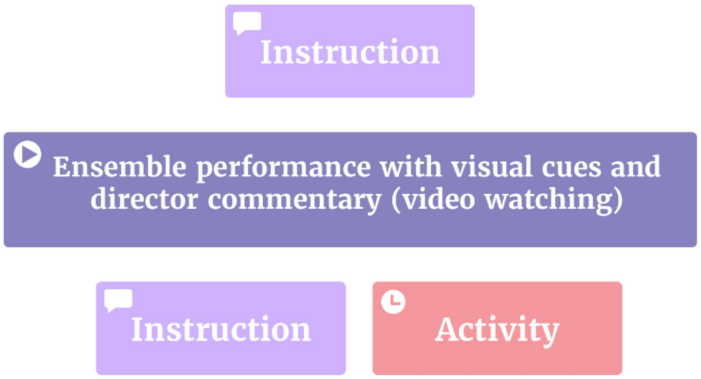


Figure 7 - Data Collection Setup

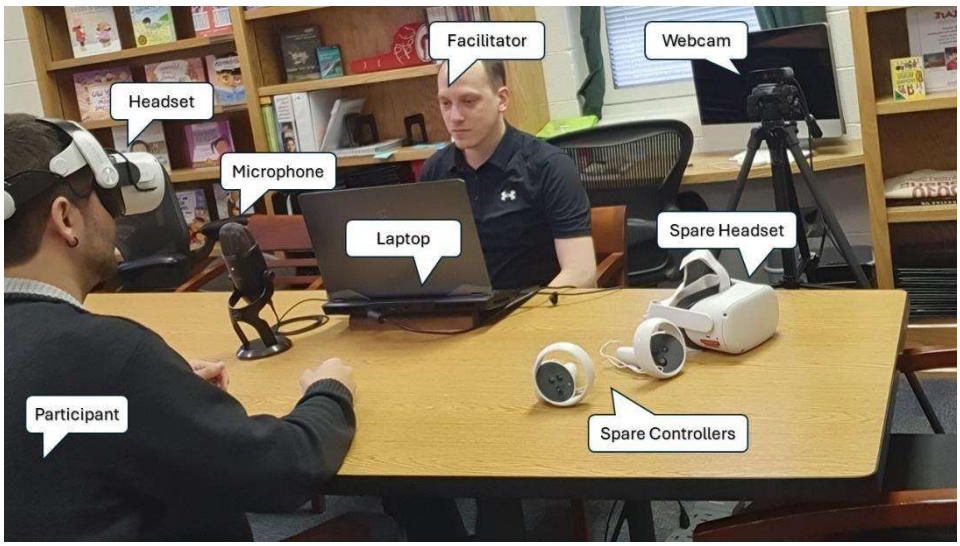


Figure 8 – Final inductive coding scheme and memo definitions for each code

CODES	MEMO DEFINITIONS
Video Segments	
Attentional	Participant was engaging in noticing, or describing things they noticed in the rehearsal environment.
Intentional	Participant described goals or acted in response to what they noticed in the rehearsal environment.
Embodied Actions	
Attentional Embodied Action	An action that gives us information about what the participant is noticing (e.g., pointing indicates a locus of attention, foot tapping indicates attention to tempo).
Intentional Embodied Action	An action taken in response to something noticed in the rehearsal environment (e.g., gestures paired with verbal feedback).
Speech	
Attentional Embodied Speech	Speech consisting of feedback given directly to the simulated ensemble concerning something the participant noticed in the rehearsal environment
Attentional Non-embodied Speech	Speech consisting of commentary concerning something the participant noticed in the rehearsal environment
Intentional Embodied Speech	Speech consisting of feedback given directly to the simulated ensemble concerning actions that could/should be taken in response to something the participant noticed in the rehearsal environment
Intentional Non-embodied Speech	Speech consisting of commentary concerning actions that could/should be taken in response to something the participant noticed in the rehearsal environment
Feedback Quality	
General	Feedback regarding general or global issues, offered to the entire simulated ensemble
Specific	Feedback targeted toward specific simulated ensemble members or musical segments